

Architecture of a Recalibration Application

20.1 Assembling the pieces

Every previous chapter built one component. This chapter assembles them into the system the book was written to explain: a recalibration application that runs, typically once per day before the desk opens (and often again intraday on demand), and turns a snapshot of the live market into a set of BGM parameters ready to price and risk-manage the book.

Key result: the recalibration pipeline, end to end

1. **Curve construction** (Chapter 2–3): bootstrap the SOFR OIS discounting curve and any required forwarding curves from live money market, futures and swap quotes.
2. **Curve validation**: check smoothness, plausibility, and absence of stale or erroneous inputs before anything downstream consumes the curve.
3. **Volatility data ingestion**: pull live cap/floor and swaption quotes (normal vols, per Chapter 1’s market convention).
4. **Caplet stripping and fit** (Chapter 16): recover spot caplet vols, fit the Rebonato parametric volatility form.
5. **Cascade calibration** (Chapter 18): fast, diagnosable sequential fit to the swaption grid, producing a starting-guess parameter set.
6. **Joint, regularized optimization** (Chapter 18): refine the cascade result against the full cap and swaption dataset simultaneously, regularized toward the previous calibration or a historical prior.
7. **Validation and reconciliation** (this chapter): reprice every input instrument with the final calibrated model and confirm residuals are within tolerance; run the ill-posedness diagnostic from Chapter 18’s worked example as standard practice, not an exception-only investigation.
8. **Publish**: write calibrated parameters to wherever the pricing and risk systems that consume them read from, with full versioning so any later price can be traced back to the exact parameter set that produced it.

20.2 Data quality: the most common real-world failure mode

Every technique in Parts III through V assumes clean, sensible input prices. In production, that assumption fails more often, and more consequentially, than any genuine model or optimizer limitation.

PITFALL

The single most common cause of an apparently “broken calibration” in practice is not the model, the correlation parameterization, or the optimizer — it is a bad input quote: a stale price that didn’t update overnight, a fat-fingered vol, a quote from a period of unusually thin liquidity that doesn’t reflect a genuinely tradable level, or a unit/convention mismatch (Chapter 1’s normal-vs-lognormal pitfall, recurring here at the system level rather than the individual-formula level). A recalibration engine that goes straight from raw market data to optimizer, with no intermediate sanity layer, will faithfully calibrate a BGM model to whatever garbage it was handed, produce a numerically converged and entirely wrong result, and give no indication anything went wrong — the most dangerous kind of failure, because it looks identical to success.

Key result: cheap checks that catch most real problems

Before any instrument feeds the optimizer, check, at minimum:

- **Cap–floor parity** (Chapter 5): an immediate, model-free consistency check on any cap/floor pair sharing a strike.
- **Monotonicity across the smile**: implied vol as a function of strike should not exhibit implausible sign flips inconsistent with the SABR-style smile shapes of Chapter 8.
- **Day-over-day jump detection**: flag any quote that moved by more than some threshold (in bp of vol) since the last successful calibration, for manual review rather than silent inclusion.
- **Cross-instrument consistency**: does the caplet-stripped vol curve (Chapter 16) look like a plausible input to the Rebonato humped form (Chapter 14), or does it require an implausible parameter combination to fit even approximately?

Each of these is inexpensive to compute relative to a full calibration run, and each catches a meaningfully different class of data problem — running all of them, every time, before the expensive optimization stage, is standard practice on any calibration system handling real risk.

20.3 How often to recalibrate, and what changes when you do**ON THE DESK**

A full recalibration — caplet strip through joint optimization — is typically run once per day, timed to run just before the desk needs live prices, precisely because it is the most computationally expensive step in the daily pipeline and its inputs (the full cap and swaption grid) only need to be captured once per trading day for most purposes. Intraday, most systems instead run a much cheaper **re-marking** step: hold the calibrated volatility and correlation *shape* parameters fixed from the morning’s full calibration, and only refresh the underlying curve (Chapters 2–3) and ATM volatility *levels* against live market moves, avoiding the cost and instability risk of a full re-optimization every time a handful of quotes tick. A genuine intraday full recalibration is usually reserved for specific triggers — a major central bank announcement, a data print that moves the curve sharply, or a risk manager’s explicit request — rather than run as a matter of routine.

20.4 Model risk and governance

A calibration engine sitting on a desk's risk numbers is, in a real sense, itself a risk — a wrong calibration produces wrong Greeks, which produces wrong hedges, which produces real trading losses regardless of how elegant the underlying mathematics is. This is not a hypothetical concern; miscalibrated term-structure models have been directly implicated in material trading losses at multiple institutions historically, which is precisely why model risk management around exactly this kind of system is now a standard, regulator-scrutinised function at any bank running one.

Key result: what a governance layer around a calibration engine needs

- **Independent validation.** A model risk function, separate from the desk that built and uses the calibration engine, periodically reviews its methodology, parameter stability, and back-tested pricing accuracy against realized market outcomes.
- **Full audit trail.** Every calibrated parameter set is versioned and timestamped, with the exact input data snapshot that produced it retained, so any historical price or risk number can be reproduced and explained after the fact.
- **Explicit limits and kill-switches.** Automated checks (the data-quality layer above, plus parameter-stability checks day-over-day) that halt automatic publication and require manual sign-off when triggered, rather than allowing a bad calibration to silently propagate into live risk numbers.
- **Documented model limitations.** A living record of exactly which products the calibrated model is validated to price (per Chapter 15's discussion of when a plain displaced-diffusion BGM suffices versus when a stochastic-volatility extension is genuinely needed), so the model is not silently relied upon outside the range it was actually built and tested for.

Worked example 20.1 — triaging a failed morning calibration.

The overnight recalibration run flags a validation failure: the 5-year cap reprices 40 bp of normal vol away from its input quote, well outside the usual sub-2 bp tolerance, while every other instrument on the grid reprices within tolerance. It is 6:45am and the desk opens at 7:30am. Walk through a fast, structured triage.

Step 1 — check the input, not the model, first. Given the error is isolated to a single instrument rather than spread across the grid (contrast with Chapter 18's ill-posedness discussion, which produces gradually growing errors across many instruments, not one sharp outlier), the prior probability strongly favours a bad input quote over a genuine model or optimizer failure. Pull the raw vendor quote for the 5-year cap and compare it to yesterday's value and to a second data vendor if one is available.

Step 2 — if the quote looks wrong. Confirm it against the day-over-day jump detector from this chapter's data-quality checklist — a 40 bp overnight jump with no corresponding move in adjacent maturities or in the swaption grid is exactly the pattern a stale-or-fat-fingered quote produces. Exclude or manually correct the single bad quote and re-run the affected stage of the pipeline (the caplet strip, since a bad flat cap vol corrupts the strip at that specific maturity and everything downstream of it) rather than the full pipeline from scratch, to stay within the time available before the desk opens.

Step 3 — if the quote looks genuine. That is a materially more serious finding — it suggests either a real, sharp market dislocation at that specific maturity (worth flagging to the

desk directly, live market information, not a data error) or a genuine limitation in the current volatility parameterization's ability to fit an unusual, newly-arrived shape. Either way, this is a case for the kill-switch: do not publish an automatically-generated fit that fails its own validation tolerance, flag it for the model risk function and the desk head, and use the previous day's calibration (clearly marked as stale) as an interim fallback.

Interpretation. The fast, correct triage path depends entirely on having the diagnostic infrastructure — isolated-versus-spread error pattern, day-over-day jump detection, a second data source, versioned prior-day fallback — already built into the system *before* the failure happens. Building this infrastructure is, in a very real sense, as much a part of “building a BGM recalibration application” as implementing Chapters 13 and 19's mathematics, and it is the part most consistently under-invested in relative to its importance.

20.5 Closing

The arc of this book runs from a single, quiet observation — rates are spreads, not prices, and can be negative — through the instruments, the history, the mathematics, and finally the calibration machinery that turns live market prices into a usable model. Every formula along the way was built to be checked by hand at least once. The practice papers that follow are designed to be worked the same way: by hand, with a calculator, checking every intermediate number, exactly as the worked examples throughout this book have modelled.

Chapter benchmark

Before moving on, you should be able to, without notes:

1. Sketch the end-to-end recalibration pipeline, in order, from curve construction to publication.
2. List at least three cheap, model-free data-quality checks that should run before any expensive calibration step, and explain what class of problem each one catches.
3. Describe, for a single bad instrument flagged by validation, a structured triage process that distinguishes a data error from a genuine model limitation, and explain why a kill-switch that blocks automatic publication on validation failure is a necessary part of the system, not an optional extra.